

Rolling-window U.S. stock returns

\$1,000 lump sum

Monthly data, Aug 1926 – Mar 2026 (1,195 months)

Generated 2026-05-16

Full-history compound annual growth rate (market)

Nominal	10.20%	[1]
Real (CPI-deflated)	6.99%	[1][2]

Mean 10-year rolling CAGR

Nominal	10.56%	[3]
Real	7.05%	[3]

Highest 10-year rolling CAGR

Nominal	+20.48%	[3]
Real	+18.12%	[3]

Lowest 10-year rolling CAGR

Nominal	-5.41%	[3]
Real	-4.97%	[3]

10-year rolling windows with CAGR ≤ 0

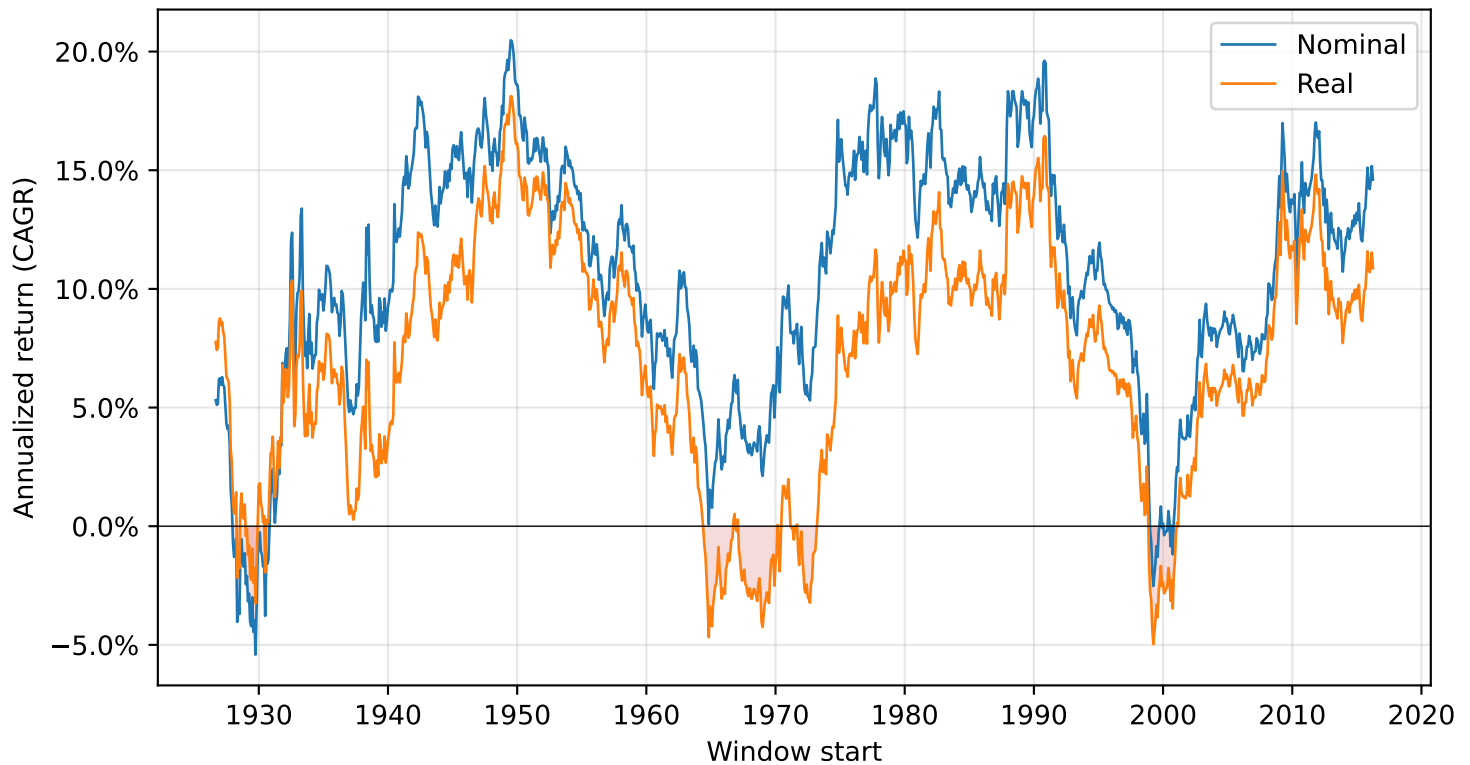
Nominal	4.65% (50 of 1,076)	[3]
Real	12.45% (134 of 1,076)	[3]

Total 10-year rolling windows	1,076	[3]
Window labeling	by start date	[3]
Total contributed per window	\$1,000	

Invest \$1,000 at the start of each rolling N-year window and hold to the end. The metric is annualized CAGR — the constant annual rate that grows \$1,000 to the terminal value over N years.

10-year rolling CAGR — Nominal & Real

Each point is the annualized return of a 10-year window beginning on that month. Red shading marks the negative region.

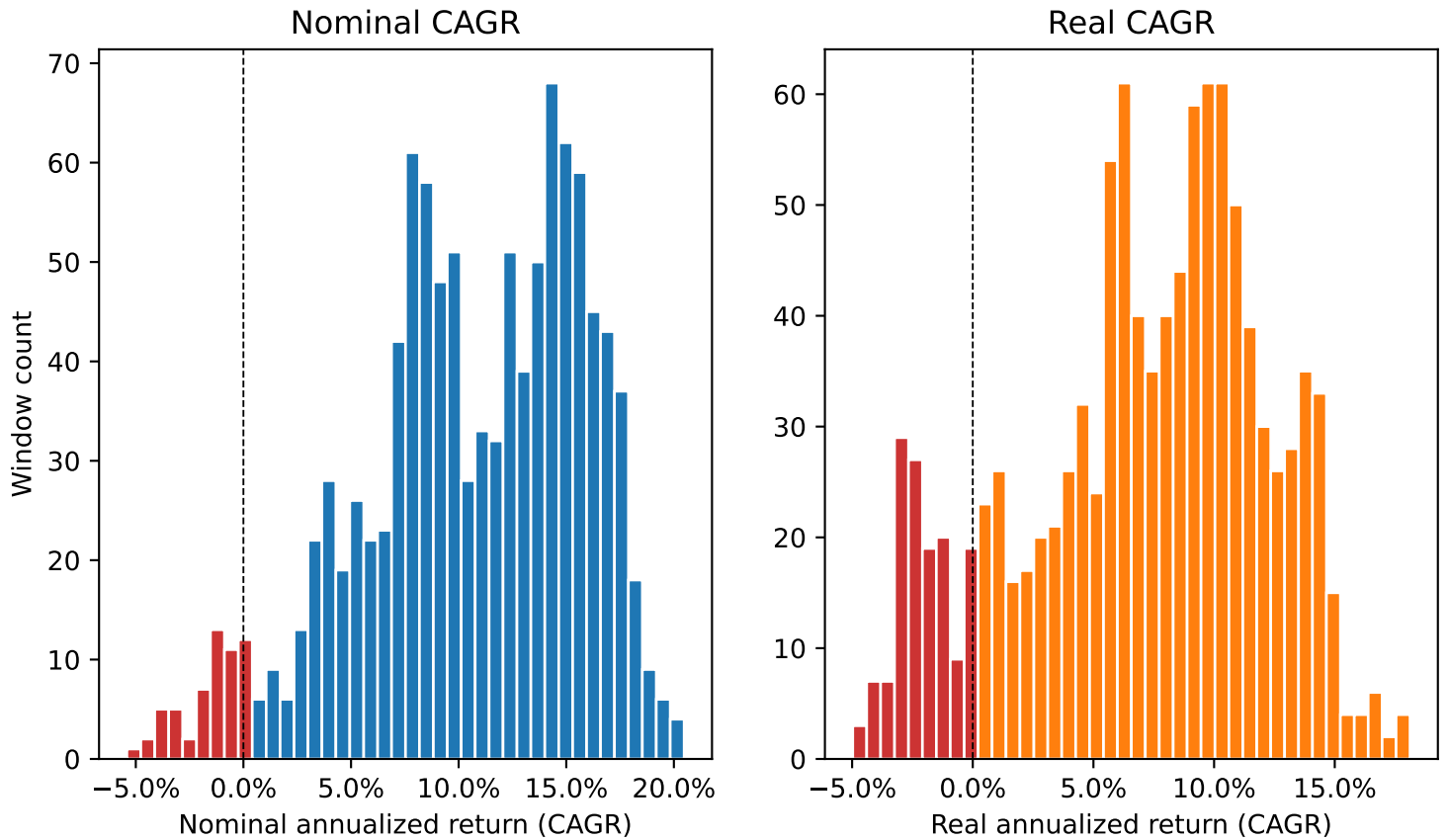


Summary statistics

	Nominal	Real
Mean	10.56%	7.05%
Median	11.10%	7.76%
Std. dev.	5.21%	5.07%
Min	-5.41%	-4.97%
Max	20.48%	18.12%
Share ≤ 0	4.65%	12.45%

Distribution of 10-year rolling CAGR — Nominal & Real

Histogram of every 10-year window in the sample. Negative bins are shown in red.



Share of windows with $\text{CAGR} \leq 0$

Nominal: 4.65% (50 of 1,076 windows)

Real: 12.45% (134 of 1,076 windows)

“ ≤ 0 ” means an investor who followed this schedule over the full 10-year window would have ended at or below their total contributions, after compounding (and after inflation, for the real series).

[3]

Best and worst 10-year windows

Top 5 highest and lowest annualized returns (CAGR), ranked by window-start month.

Best nominal

Start	End	CAGR	\$1,000 →
Jun 1949	May 1959	20.48%	\$6,442
Jul 1949	Jun 1959	20.45%	\$6,429
Aug 1949	Jul 1959	20.20%	\$6,295
May 1949	Apr 1959	19.89%	\$6,137
Sep 1949	Aug 1959	19.74%	\$6,058

Worst nominal

Start	End	CAGR	\$1,000 →
Sep 1929	Aug 1939	-5.41%	\$573
Jul 1929	Jun 1939	-4.45%	\$634
May 1929	Apr 1939	-4.22%	\$650
Apr 1928	Mar 1938	-4.03%	\$662
Apr 1929	Mar 1939	-4.03%	\$663

Best real

Start	End	CAGR	\$1,000 →
Jun 1949	May 1959	18.12%	\$5,287
Jul 1949	Jun 1959	18.10%	\$5,280
Aug 1949	Jul 1959	17.72%	\$5,109
May 1949	Apr 1959	17.60%	\$5,057
Mar 1949	Feb 1959	17.35%	\$4,953

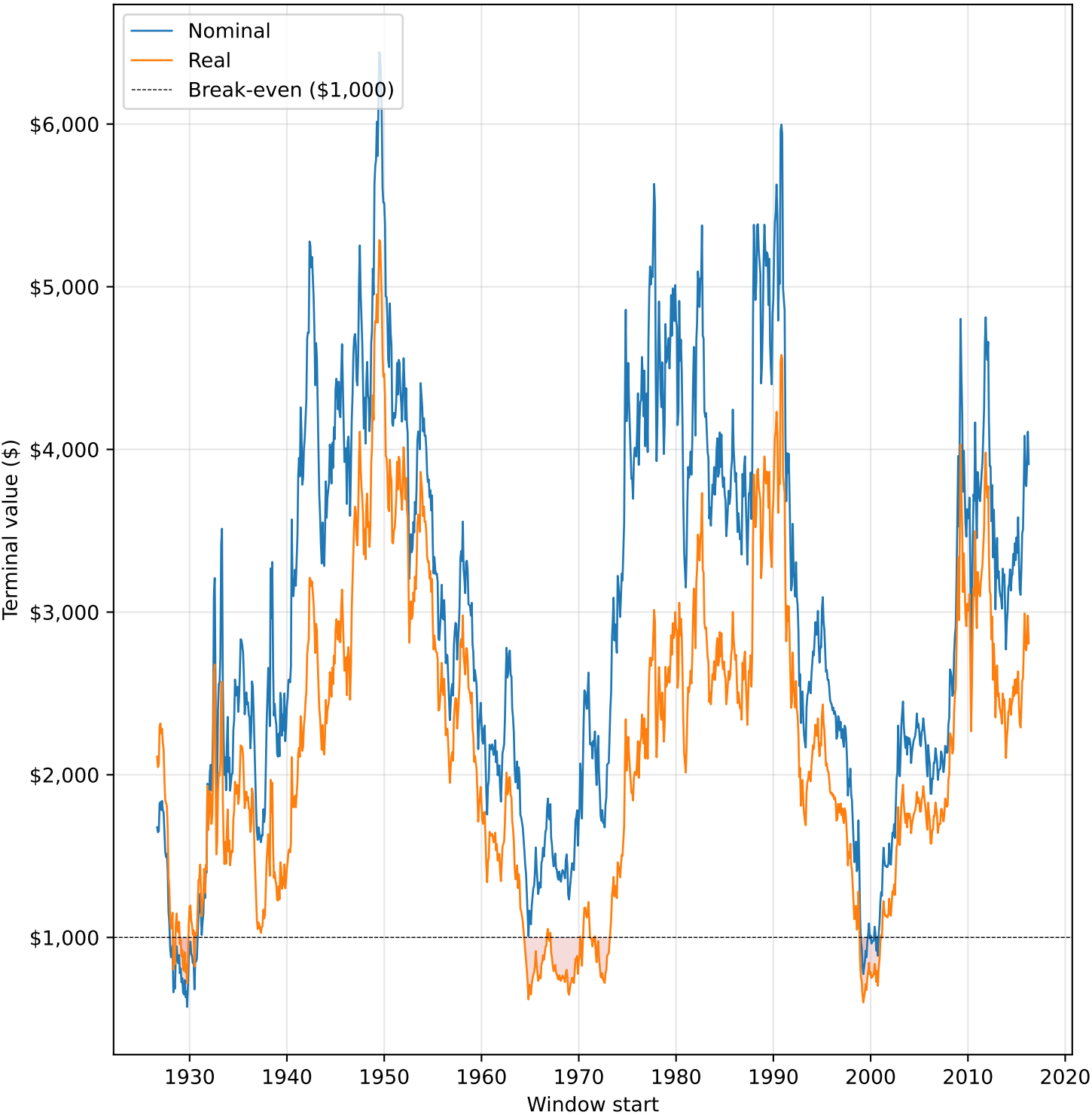
Worst real

Start	End	CAGR	\$1,000 →
Mar 1999	Feb 2009	-4.97%	\$600
Oct 1964	Sep 1974	-4.68%	\$619
Apr 1999	Mar 2009	-4.51%	\$630
Feb 1999	Jan 2009	-4.28%	\$646
Dec 1968	Nov 1978	-4.25%	\$648

inal value of the full contribution schedule, held to the end of the 10-year window. [4] Red marks a negative CAGR or a terminal below t

Terminal value of \$1,000 after 10 years — Nominal & Real

re \$1,000 contributed under this schedule would have landed, by start month. Red shading marks where the terminal fell below break-e



Sources and methodology

Data sources

- [1] Fama/French Research Data Factors, monthly file. Ken French Data Library, Tuck School of Business, Dartmouth.
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Columns used: Mkt-RF (market excess return), RF (risk-free rate).
Values are published as percent (2.50 = 2.5%) and divided by 100 on ingest.
- [2] U.S. CPI for All Urban Consumers, not seasonally adjusted (CPIAUCNS).
Federal Reserve Bank of St. Louis (FRED).
<https://fred.stlouisfed.org/series/CPIAUCNS>
The not-seasonally-adjusted series is used deliberately – the SA series (CPIAUCSL) only begins in 1947 and would truncate 21 years of history.

Series construction

```
nominal_return = (Mkt-RF + RF) / 100 - total U.S. market return.  
inflation      = CPI.pct_change() - month-over-month CPI change.  
real_return    = (1 + nominal) / (1 + inflation) - 1 (Fisher equation).  
The first observation (Jul 1926) is dropped because monthly inflation  
requires a prior CPI reading.
```

[3] Rolling 10-year window math — \$1,000 lump sum

For each month t , compound the next $N*12$ monthly returns:

```
terminal(t) = $1,000 * exp(  $\sum \log(1 + r_i)$  ) for  $i$  in  $[t, t+N*12-1]$   
CAGR(t)     = (terminal(t) / $1,000) ^ (1 / N) - 1
```

Log-returns are summed (not products chained) for numerical stability over long horizons. Each window is labeled by its START month, not its end. Windows lacking a full $N*12$ months are dropped from the tail.

[4] Full-history market CAGR

```
terminal = exp(  $\sum \log(1 + r_t)$  ) over the full sample of monthly returns.  
CAGR     = terminal ^ (12 / n_months) - 1.  
Shown on the cover for context; independent of the scenario schedule.
```

Reproducibility

```
python -m src.ingest      # rebuild data/processed/monthly_returns.parquet  
python -m src.export      # rebuild docs/data/*.json  
python -m src.report      # rebuild docs/reports/*.pdf  
Sample window: Aug 1926 – Mar 2026 (1,195 monthly observations).
```